TEXAS STAAR

GRADE 8

Reading Language Arts & Mathematics

All-in-One Study & Practice Guide

8 Full-Length Practice Tests · Complete Skill Review · Deep Answer Explanations

Short & Extended Constructed Response Mastery · Score Diagnostics

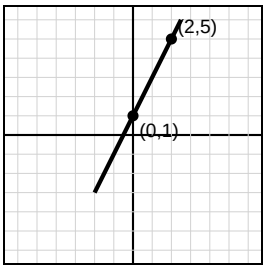
100% aligned to the Texas Essential Knowledge and Skills (TEKS)

2027 EDITION

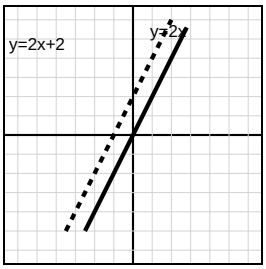
Andrian Ross

Strand 2 · Computations and Algebraic Relationships

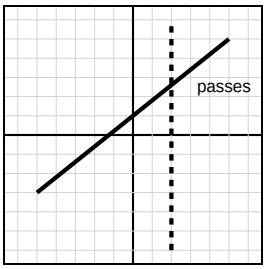
Key Figures for This Strand



Slope from a graph (8.4C)



Proportional vs. not (8.5A/B)



Vertical line test (8.5G)

This is the **largest strand** on the test (about 15 of 40 questions). It is all about **lines, relationships, and equations.** Master it and you master most of the Math test.

Standards covered:

- ○ **8.4A** — Similar right triangles show slope is the same between any two points on a line.
- ★ **8.4B** — Graph proportional relationships; interpret the unit rate as the slope.
- ★ **8.4C** — Use a table or graph to find the rate of change (slope) and the y-intercept.
- ○ **8.5A** — Linear proportional situations as y = kx. ○ **8.5B** — Non-proportional as y = mx + b (b ≠ 0).
- ○ **8.5E** — Direct variation. ○ **8.5F** — Distinguish proportional from non-proportional. ○ **8.5H** — Examples of each.
- ★ **8.5G** — Identify functions from ordered pairs, tables, mappings, graphs.
- ★ **8.5I** — Write an equation y = mx + b to model a linear relationship.
- ○ **8.8A** — Write one-variable equations/inequalities with variables on both sides. ○ **8.8B** — Write a real-world problem for a given equation. ★ **8.8C** — Model and solve them.
- ○ **8.9A** — Identify a system's solution from the intersection of two graphs.

Part A · Slope and Rate of Change (8.4A ○, 8.4B ★, 8.4C ★)

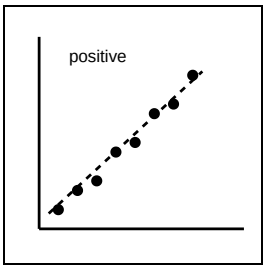
Slope measures how steep a line is — how much *y* changes for each change in *x*:

$$\textbf{slope} = m = (\text{change in } y) \div (\text{change in } x) = (y_2 - y_1) / (x_2 - x_1)$$

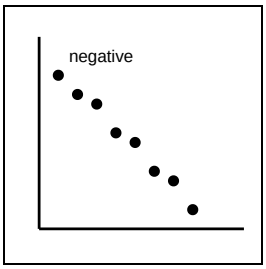
- **8.4A — Why slope is constant.** Right triangles drawn to any two points on the same line are **similar**, so rise ÷ run is the **same everywhere** — that ratio is the slope.

Strand 4 · Data Analysis and Personal Financial Literacy

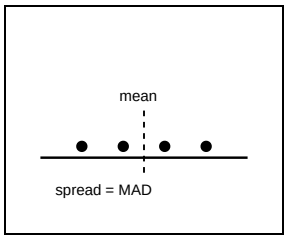
Key Figures for This Strand



Positive trend (8.5D)



Negative association (8.11A)



Mean absolute deviation (8.11B)

This strand (about 8 of 40 questions) has two halves: **reading data** (scatterplots, trend lines, spread) and **personal finance** (interest, saving, the cost of borrowing).

Standards covered:

- ○ **8.5C** — Contrast bivariate data that suggest a linear relationship with data that do not. ★ **8.5D** — Use a trend line to make predictions.
- ○ **8.11A** — Construct a scatterplot and describe the association. ○ **8.11B** — Mean absolute deviation (≤10 points).
- ○ **8.12A** — How rate and loan length affect the cost of credit. ○ **8.12C** — How regular saving grows. ★ **8.12D** — Simple vs. compound interest. ○ **8.12G** — Estimate college cost and a savings plan.

Part A · Scatterplots and Association (8.5C ○, 8.11A ○)

A **scatterplot** graphs paired (bivariate) data. The pattern shows the **association**: **positive linear** (points rise), **negative linear** (points fall), **non-linear** (a curve), or **no association** (scattered).

Worked example. Study time up, scores up → points rise right → **positive linear.** **Try It.** TV hours up, grades down → *negative linear.*

Part B · Trend Lines and Predictions (8.5D ★)

A **trend line** (line of best fit) runs through the middle of the data; use it to **predict** values you did not measure.

Worked example. Trend line y = 2x + 3 (x = weeks). Height at week 10: y = 2(10) + 3 = **23 cm.** **Watch Out:** a prediction is an **estimate**, not exact.